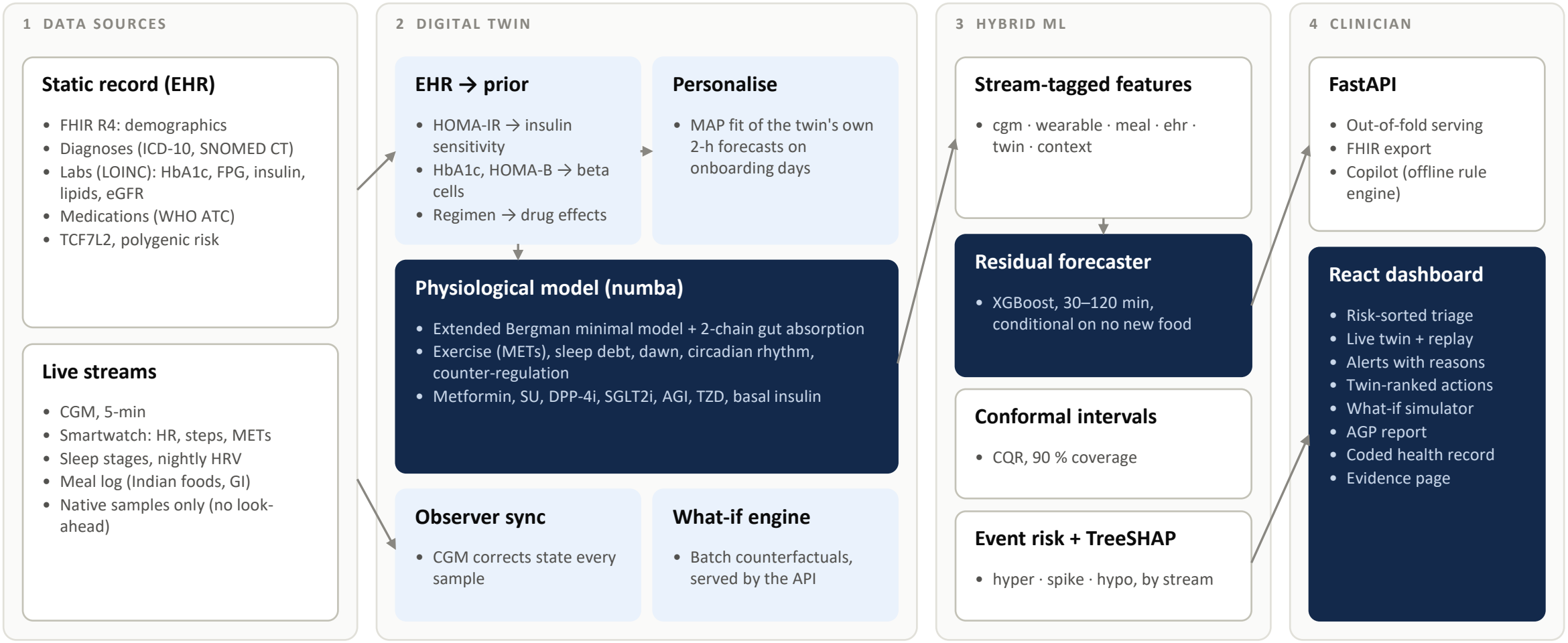
System architecture

Two data streams in, a personalised physiological twin plus a hybrid ML layer in the middle, decisions for the clinician out.



Governance: DPDP-aligned consent per patient · decision support (no dosing) · synthetic data always labelled · every on-screen prediction from a model that never saw that patient

Validation and deployment pipeline

Every number in the README, slides and dashboard is regenerated by these scripts from public or synthetic data.

REAL-DATA EVIDENCE (CGMACROS, PHYSIONET, 45 ADULTS)

download_cgmacros.py
- S3 mirror, SHA-256 check

→ **Causal loader**
- Native sensor samples
- METs from calories

→ **Per-patient twin**
- Onboard: first 40 % (≤4 d)
- Monitor: the rest

→ **5-fold leave-patients-out**
- Stratified by glycaemic group
- Same rows for every model

→ **Metrics**
- RMSE, MARD, Clarke grid
- AUROC, lead time, coverage
- Meal response, ablation

DEPLOYMENT (BHARAT SYNTHETIC COHORT, 80 FICTIONAL INDIAN PATIENTS)

build_bharat.py
- Physiology-first generator
- Indian diets, drugs, stories

→ **Twin + features**
- Same pipeline as real data

→ **train_deploy.py**
- 5 fold models: all CGMacros
- + 4/5 of the Bharat cohort

→ **Out-of-fold predictions**
- Each patient scored by the
- model that never saw them

→ **API + dashboard**
- Replay clock, alerts, what-if
- Synthetic metrics labelled

REPRODUCE, IN ORDER

```
scripts/  download_cgmacros → build_cgmacros → evaluate_cgmacros → build_bharat → train_deploy   then   uvicorn madhutwin.api.main:app
```

Tests: physiology sanity, a look-ahead test (features at minute t are identical whether or not later data exists), metric checks, end-to-end API smoke tests.